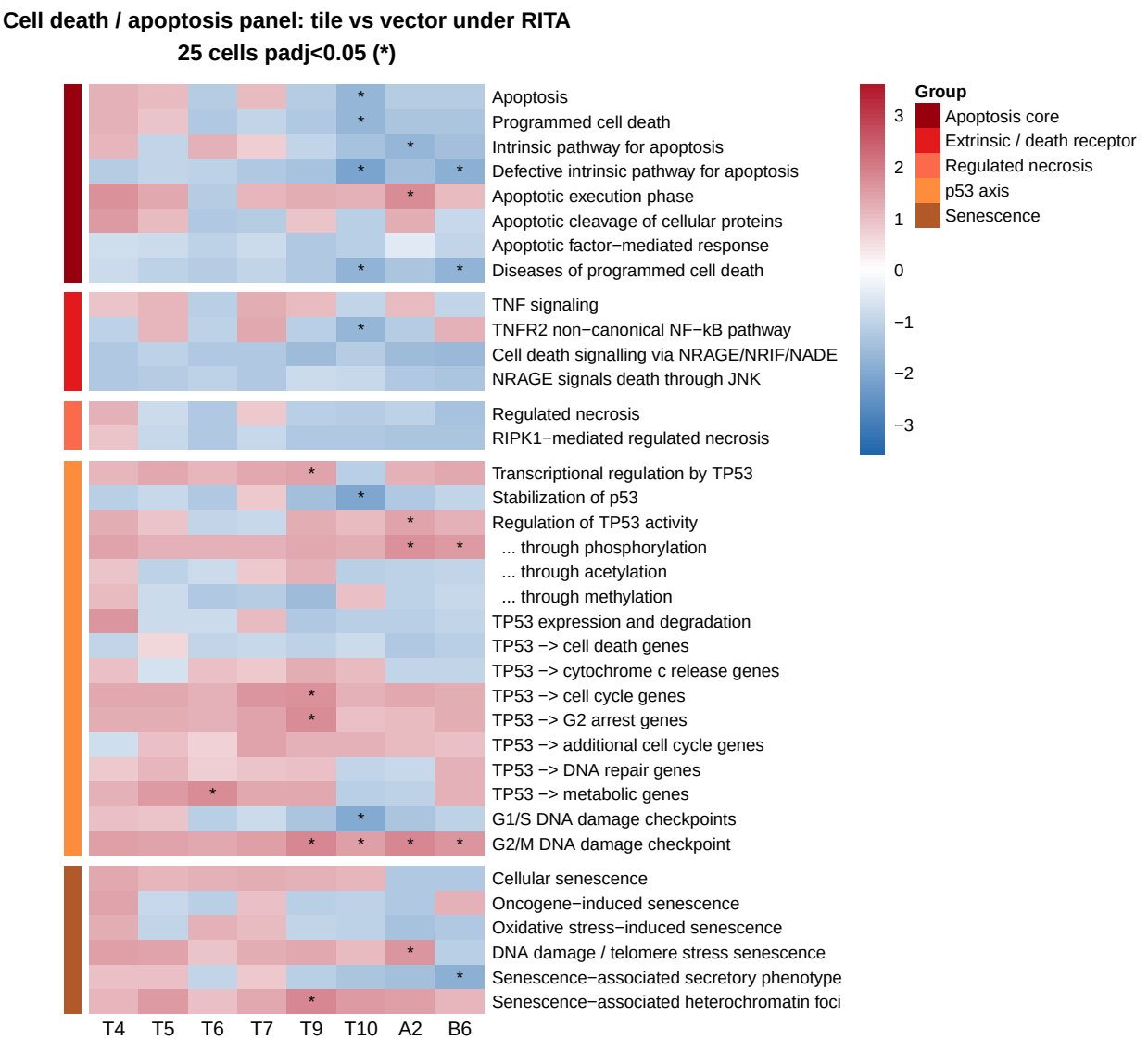
Cell death / apoptosis panel: tile vs vector under RITA
25 cells padj<0.05 (*)
Apoptosis
Programmed cell death
Intrinsic pathway for apoptosis
Defective intrinsic pathway for apoptosis
Apoptotic execution phase
Apoptotic cleavage of cellular proteins
Apoptotic factor−mediated response
Diseases of programmed cell death
TNF signaling
TNFR2 non−canonical NF−kB pathway
Cell death signalling via NRAGE/NRIF/NADE
NRAGE signals death through JNK
Regulated necrosis
RIPK1−mediated regulated necrosis
Transcriptional regulation by TP53
Stabilization of p53
Regulation of TP53 activity
... through phosphorylation
... through acetylation
... through methylation
TP53 expression and degradation
TP53 −> cell death genes
TP53 −> cytochrome c release genes
TP53 −> cell cycle genes
TP53 −> G2 arrest genes
TP53 −> additional cell cycle genes
TP53 −> DNA repair genes
TP53 −> metabolic genes
G1/S DNA damage checkpoints
G2/M DNA damage checkpoint
Cellular senescence
Oncogene−induced senescence
Oxidative stress−induced senescence
DNA damage / telomere stress senescence
Senescence−associated secretory phenotype
Senescence−associated heterochromatin foci
T4
T5
T6
T7
T9
T10
A2
B6
Group
Apoptosis core
Extrinsic / death receptor
Regulated necrosis
p53 axis
Senescence
3
2
1
0
−1
−2
−3